

SVM in ML

- 1) Support vector machine (SVM) → Support Vector Machine (SVM) is a supervised machine learning algorithm mainly used for classification and regression.

It finds best decision boundary (hyperplane) that separates different classes with the maximum possible margin.

2) Imp terms in SVM.

i) Hyperplane: It is a decision boundary that separates classes. It uses line for 2D data.

ii) Margin: The distance between decision boundary and nearest point ~~to~~ from each class.
larger margin → better separation

iii) Support Vectors: The closest data points to the decision boundary. It is important to determine position of the boundary.

iv) Kernel: A technique used when data cannot be separated by a straight line.

either: linear

sigmoid

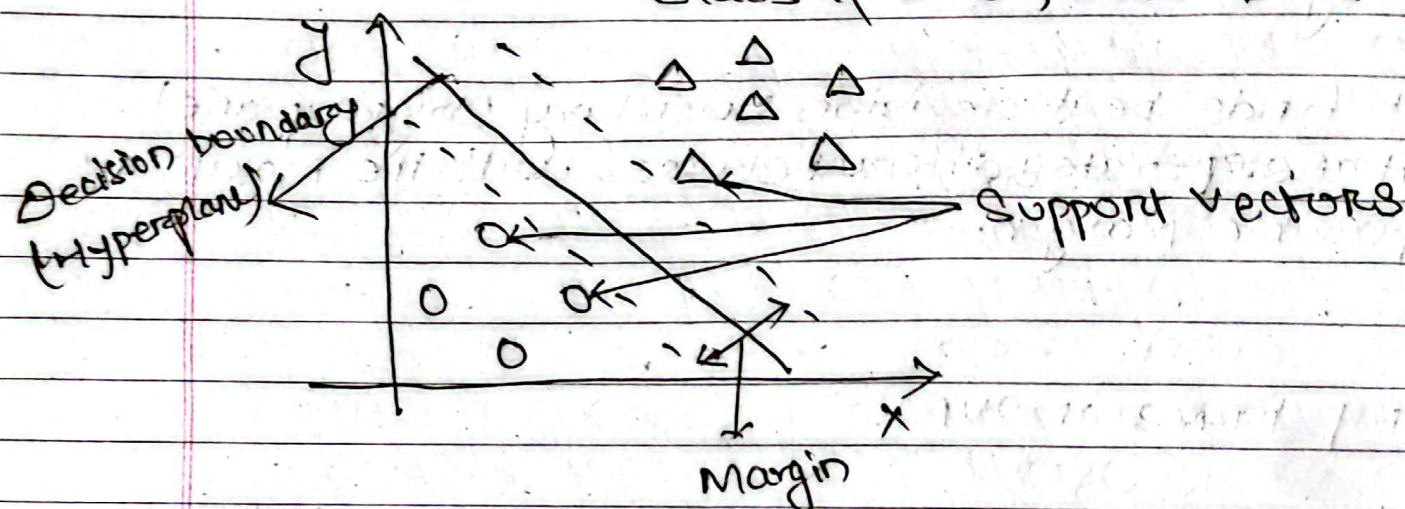
polynomial

RBF

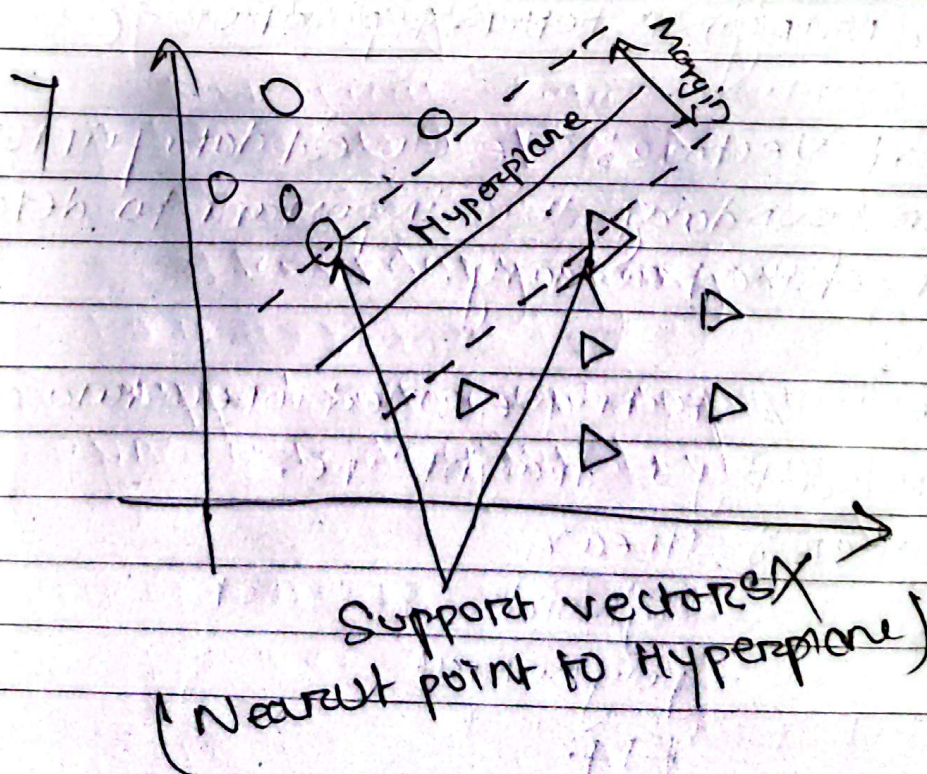
3) How SVM works?

Step 1: SVM searches for a line/hyperplane that separate two classes.

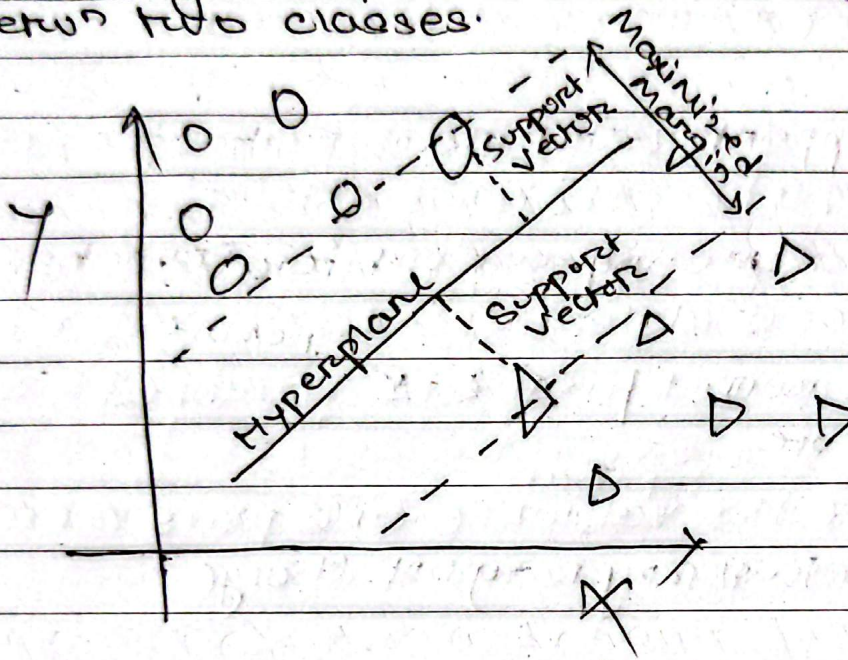
Class A = Δ , class B = $\circ$



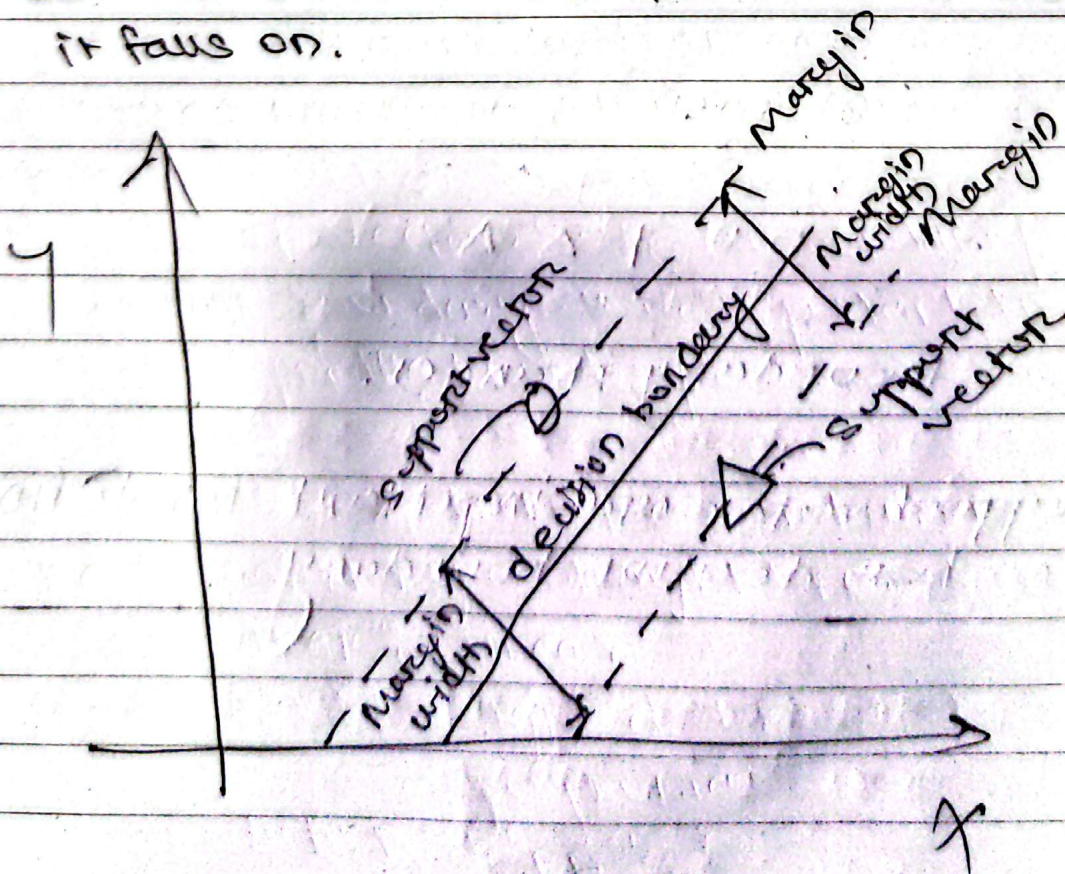
Step 2: Find the nearest points (closest point to the ~~boundary~~) decision boundary). These will be support vectors.



Step 3: Maximize the margin by choosing the boundary that provides the largest margin between two classes.



Step 4: Classify new data, when a new data point arrives SVM checks which side of the boundary it falls on.

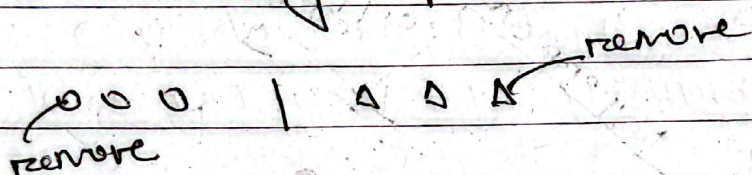


Which side the margin width is more the new data fall on.

5. Why support vector is important?

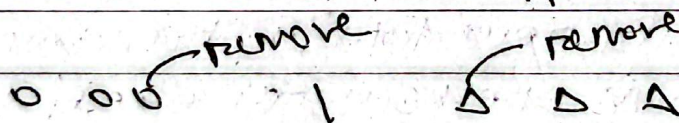
Imagine removing a point far from boundary.

eg:



Boundary may not change

But if we remove a support vector



change



Boundary changes.

∴ support vector are important for defining SVM ~~but~~ decision boundary.